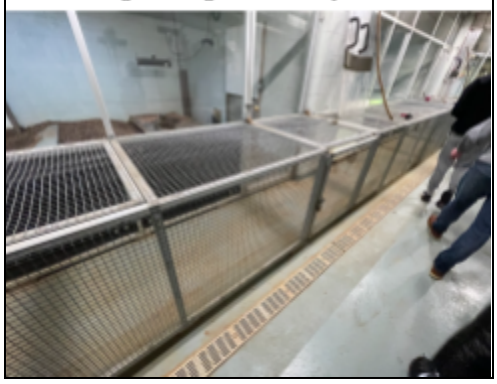
Komodo Dragon Element A

Zoos across the nation are a vital part of preserving the biodiversity and beauty of nature. Through hundreds of years of research and experience, zoos and their staff have learned to house and care for a wide variety of creatures, from giraffes to fish. However, some species are relatively rare and zoos have limited experience caring for them.

One such example is the Komodo dragon. With a population of around 3500 individuals worldwide[5], researchers and zookeepers have had relatively limited experience caring for these dragons. These unique creatures are the largest lizards in the world, with venomous bites, chainmail-like skin, razor-sharp claws and teeth, and extremely mobile bodies[1]. These qualities that intrigue zoo-goers also present a unique challenge in safely handling them.

Currently, our local zoo, the Nashville Zoo at Grassmere, has the largest Komodo dragon habitat in North America[6], with one adult male, two adult females, two juvenile males, and two hatchlings[7]. After interviewing two of our primary stakeholders -- Nick Hanna, Assistant Curator of Ectotherms, and James Flaherty, Herpetology Keeper -- we learned that one of the main problems they face regularly was transporting the dragons from building to building.

Fig 1 - Dragon Tunnel System



Their exhibit has a system of metal tunnels (see Fig 1) to move the dragons between the outdoor space, indoor space, and individual habitats that works well but is not linked up to the HCA (the animal hospital at the zoo). Whenever they need to transport the dragons to the HCA or beyond, they currently use these black pipes (see Fig 2) with cutouts for a solid polypropylene gate that slots in at both ends. While this solution is primitive, it can effectively hold the dragons. The tight dimensions help limit the dragons' ability to use their tails and claws, the darkness acts as a calming agent, and the guillotine-style sheets allow the keepers to close the gates without putting their hands near the dragon's tail or teeth.

Fig 2 - Current Transport Solution

However, the keepers and hospital technicians have expressed safety concerns with the current solution, especially regarding visibility and access to the dragons while in the tube. One main issue is not being able to see where the dragon is along the tube and if the dragon has twisted or turned around. Objects used to currently keep the zookeepers safe are police shields and y shaped sticks. This system is not effective because the dragons can easily make their way around the safety mechanism if misused by the keeper. Although these products are upholding the safety of the zookeepers now, a higher-grade safety system would be more appropriate.

Another concern is that X-rays and ultrasounds cannot be used through the tubes. The ultrasounds and x-rays have to be placed under or on the dragon's stomach to be effectively used and to obtain clear pictures. For example, when we visited the zoo animal hospital, the

veterinarians explained how a box with a place to slide the X-ray plate into would be very helpful for them. They also explained how X-ray pictures do not come out clear when shot through metal or similar materials, so they asked that our box be made of a material they could work with.

Additionally, the zookeepers asked for the box to be easy to hose out, have handles on the ends, and fit through the doorway of the komodo's exhibit, which we measured to be approximately 24" wide.

The primary stakeholders for this project include the komodo dragon, the komodo dragon handlers, the veterinarian staff, and the zoo administration. Secondary stakeholders include zoo attendees and other zoos across the world.

Works Cited

[1] Smithsonian's National Zoo & Conservation Biology Institute. "Komodo Dragon." Smithsonian's National Zoo, 9 July 2018, nationalzoo.si.edu/animals/komodo-dragon.

[2] Dragon, Komodo. "Animal Training." Animal Training, 2025, www.animaltraining.org/komodo-dragon/.

[3] "Nashville Zoo Paving the Way for Komodo Dragon Reproduction Research." Www.nashvillezoo.org, www.nashvillezoo.org/our-blog/posts/nashville-zoo-paving-the-way-for-komodo-dragon-reproduction-research.

[4] Fleming, Gregory. HUSBANDRY and MEDICAL MANAGEMENT of KOMODO DRAGONS (Varanus Komodoensis) at the WHITE OAK CONSERVATION CENTRE. 1997.

[5] Jirik, Kate. "LibGuides: Komodo Dragon (Varanus Komodoensis) Fact Sheet: Summary." Ielc.libguides.com, ielc.libguides.com/sdzg/factsheets/komododragon.

[6] "Largest Komodo Dragon Habitat in the Americas at Nashville Zoo." Www.aza.org, www.aza.org/connect-stories/stories/largest-komodo-dragon-habitat-in-the-americas-at-nashville-zoo.

[7] "Year of the Dragon: Nashville Zoo Welcomes Komodo Hatchlings." Nashvillezoo.org, 2024, www.nashvillezoo.org/our-blog/posts/nashville-zoo-welcomes-komodo-hatchlings.

[8] Hanna, Nick. Personal interview. 17 Jan. 2025.

[9] Grassmere Zoo Staff. Personal interview and tour. 30 Jan. 2025.